

Given: $E = 1.781 \times 10^6 \text{ psi}$, $S_{\text{MOR}} = 14080 \text{ psi}$, $W = 71.25 \text{ lbf}$, $L_e = 60.5 \text{ in}$, $d = 1.25 \text{ in}$

$$w = \frac{W}{L_e} = \frac{71.25 \text{ lbf}}{60.5 \text{ in}} = 1.186 \text{ lbf/in} \quad \text{Distributed load} \quad R_A = R_B = \frac{W}{2} = \frac{71.25 \text{ lbf}}{2} = 35.875 \text{ lbf} \quad \text{Reaction Force at each end}$$

$$I = \frac{\pi d^4}{64} = \frac{\pi (1.25 \text{ in})^4}{64} = 0.1198 \text{ in}^4 \quad M_{\text{max}} = \frac{w L_e^2}{8} = \frac{W L_e}{8} = \frac{(71.25 \text{ lbf})(60.5 \text{ in})}{8} = 542.6 \text{ lbf}\cdot\text{in} \quad \text{Maximum moment at midspan}$$

$$\sigma_{\text{max}} = \frac{M_{\text{max}} c}{I} = \frac{M_{\text{max}} d}{2I} = \frac{(542.6 \text{ lbf}\cdot\text{in})(1.25 \text{ in})}{2(0.1198 \text{ in}^4)} = 2830 \text{ psi} = 2.830 \text{ ksi}; \text{ Maximum Bending Stress}$$

$$FS_{\text{MOR}} = n_{\text{MOR}} = \frac{S_{\text{MOR}}}{\sigma_{\text{max}}} = \frac{14080 \text{ psi}}{2830 \text{ psi}} = 4.98; \text{ Factor of safety between modulus of rupture, which can be used for wood bending failure, and the maximum experienced bending stress}$$

$$\delta_{\text{max}} = \frac{5 w L_e^4}{384 E I} = \frac{5 (1.186 \text{ lbf/in})(60.5 \text{ in})^4}{384 (1.781 \times 10^6 \text{ psi})(0.1198 \text{ in}^4)} = 0.969 \text{ in}; \text{ maximum deflection of rod under experienced loading}$$

$$\delta_{\text{allow}} = \frac{L_e}{240} = \frac{60.5 \text{ in}}{240} = 0.252 \text{ in} \quad \frac{\delta_{\text{max}}}{\delta_{\text{allow}}} = \frac{0.969 \text{ in}}{0.252 \text{ in}} = 3.84, \delta_{\text{max}} > \delta_{\text{allow}}; \text{ Sag is more critical than failure using } \frac{L}{240} \text{ serviceability values}$$

$$W_{\text{rupture}} = \frac{8 S_{\text{max}} I}{c L_e} = \frac{8 (14080 \text{ psi})(0.1198 \text{ in}^4)}{(0.625 \text{ in})(60.5 \text{ in})} = 357 \text{ lbf}; \text{ Load required to rupture rod}$$

$$W_{L/240} = W \left(\frac{\delta_{\text{allow}}}{\delta_{\text{max}}} \right) = 71.25 \text{ lbf} \left(\frac{0.252 \text{ in}}{0.969 \text{ in}} \right) = 18.7 \text{ lbf}; \text{ Load required to push rod past serviceability limit and into sag criterion}$$

$$\delta_{\text{creep}} \approx 1.5 \delta_{\text{max}} = 1.5 (0.969 \text{ in}) = 1.45 \text{ in}; \text{ simplified long-term deflection estimate}$$

Key Values:

$w = 1.186 \text{ lbf/in}$	$\sigma_{\text{max}} = 2.830 \text{ ksi}$	$\delta_{\text{max}} = 0.969 \text{ in}$
$R_A = R_B = 35.875 \text{ lbf}$	$FS_{\text{MOR}} = 4.98$	$\delta_{\text{allow}} = 0.252 \text{ in}$
$M_{\text{max}} = 542.6 \text{ lbf}\cdot\text{in}$		

Sources / Reasonings:

- S_{MOR} , modulus of rupture, is described by the Wood Database as an alternative ultimate/yield strength for wood because it behaves differently than steel in bending.
- Area moment of inertia, I , is for a solid cylinder in one-directional bending.
- Maximum moment and deflection, M_{max} and δ_{max} , are previously derived equations for simply supported beams experiencing a uniform distributed load.
- Allowable estimates ($\frac{L_e}{240}$ serviceability) are experimentally derived from the American Wood Council.